

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	14 February 2025
Team ID	LTVIP2026TMIDS54574
Project Name	Smart Sorting Transfer Learning for Identifying Rotten Fruits and Vegetables
Maximum Marks	4 Marks

Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Image Upload	User uploads fruit or vegetable image through web interface
FR-2	Image Preprocessing	System resizes, normalizes, and prepares image for prediction
FR-3	Rotten/Fresh Prediction	System classifies image as Fresh or Rotten using trained ResNet18 model
FR-4	Display Prediction Result	System displays prediction result with confidence percentage
FR-5	Model Integration	System loads trained deep learning model for inference
FR-6	File Handling	System stores uploaded images temporarily for processing
FR-7	User Interaction	System provides simple and user-friendly web interface

Non-functional Requirements:

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system must provide an easy-to-use interface for image upload and prediction
NFR-2	Security	Uploaded files must be handled securely without exposing system vulnerabilities
NFR-3	Reliability	System must provide consistent and accurate predictions
NFR-4	Performance	Prediction should be generated within a few seconds
NFR-5	Availability	System should be available whenever the server is running

NFR-6	Scalability	System can be deployed on cloud platforms to handle multiple users
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